UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Important points

Definitions, statements, key points, "why" reasoning and exam traps, topic by topic.

Class 011 (VoP) · Method of variation of parameters · compiled 16 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S011V·9	Unit 2 deck 011 "Variation of Parameters method" (SESSION 8), slide 9 (rendered .pptx)
T011V 00:03:26	AV-summary transcript "Variation of Parameters method" (14.9 min, listed first), segment start
T011Vb 00:16:53	AV-summary transcript "Variation of Parameters method_Continued" (24.5 min, second), segment start
CW U2 P25	Unit 1 & 2 class-work sheet, Unit 2 problem 25
ESA Jul-23 3c	Past paper: UE22MA141A July 2023, question 3.c (Jan-24 s1/s2 = UE23MA141A Jan 2024 sets 1/2)

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1 The method: $y_p = uy_1 + vy_2$ with the Wronskian

Definitions

Method of variation of parameters. A method to find the particular integral of a linear differential equation whose complementary function is known, by replacing the constants c_1, c_2 of $y_c = c_1y_1 + c_2y_2$ by functions $u(x)$, $v(x)$. S011V·4 · T011V 00:00:16 · T011V 00:00:25

Wronskian. $W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1y_2' - y_2y_1'$; it is non-zero when y_1, y_2 are linearly independent solutions (class 001). S011V·5 · T011V 00:02:35 · T011V 00:03:05

Linearly independent. y_1, y_2 are linearly independent if $c_1y_1 + c_2y_2 \equiv 0$ only for $c_1 = c_2 = 0$; then $c_1y_1 + c_2y_2$ is the CF. (added for clarity) T011V 00:01:48 · T011V 00:02:06

Standard form. $\frac{d^2y}{dx^2} + p\frac{dy}{dx} + qy = X$: the coefficient of $\frac{d^2y}{dx^2}$ is 1 and X is the right-hand side of this form (divide the whole equation by the leading coefficient first). S011V·4 · S011V·6 · T011V 00:06:06

Parameters u, v. The functions $u = -\int \frac{y_2X}{W}dx$ and $v = \int \frac{y_1X}{W}dx$, written without constants of integration. S011V·5 · T011V 00:03:26 · T011V 00:04:18

Statements in the class material

> *"This method is used to find the particular integral of a linear differential equation whose complimentary function is known. Though this method is general, we will illustrate it by applying to a second order LDE with constant coefficients. Consider a second order linear differential equation* $\frac{d^2y}{dx^2} + p\frac{dy}{dx} + qy = X$ *where* p *and* q *are constants and* X*, a function of* x*. Let its complimentary function be* $CF = c_1y_1 + c_2y_2$*."* S011V·4

> *"*W *is the wronskian of* y_1 *and* y_2 *and is given by* $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$*. The particular solution of the differential eqn is given by* $PI = u.y_1 + v.y_2$ *where* u *and* v *are given by* $u = -\int \frac{y_2.X}{W}dx$ *and* $v = \int \frac{y_1.X}{W}dx$*. Note: The constant of integration is not written in the expressions of* u *and* v*."* S011V·5

> *"This method can also be used on differential equation with variable coefficient provided we know to find the general solution of the associated homogeneous equation. The coefficient of* $\frac{d^2y}{dx^2}$ *in the differential equation must be unity. If it is not unity, make it one by dividing the differential equation by the coefficient."* S011V·6

Key points

1. The method gives only the particular integral, and it needs the complementary function first: "the complementary function is absolutely required to apply this method". ESA ★★★ S011V·4 · T011V 00:00:25 · T011V 00:06:42 · T011Vb 00:01:12
2. Three steps: (1) CF, read off y_1, y_2; (2) Wronskian $W = y_1y_2' - y_2y_1'$; (3) $u = -\int \frac{y_2X}{W}dx$, $v = \int \frac{y_1X}{W}dx$, $y_p = uy_1 + vy_2$ (Figure F1.1). ESA ★★★ T011V 00:08:55 · T011Vb 00:01:29 · T011Vb 00:08:20 · S011V·7 · S011V·8 · S011V·9
3. Make the coefficient of $\frac{d^2y}{dx^2}$ equal to 1 before reading off X; otherwise u and v are wrong by that factor. ESA ★★★ S011V·6 · T011V 00:06:06 · CW U2 P27
4. No constants of integration in u, v: "the constant of integration gets absorbed with the constants of the complementary function". ESA ★★★ S011V·5 · T011V 00:04:18 · T011Vb 00:07:06
5. Use it when the question says "by the method of variation of parameters" (every ESA VoP question does), or when X is not a standard form ($\tan$, $\sec$, $\csc$, $\frac{e^{mx}}{x}$, $\frac{1}{1+e^x}$, ...). ESA ★★★ T011V 00:08:19 · ESA Jul-23 3c · ESA Jan-24 s1 3a · CW U2 P23
6. It is general: it also works for variable coefficients when the CF is known, and for higher order; only second order is taught. ESA ★ S011V·6 · T011V 00:05:09 · T011V 00:05:39 · T011Vb 00:16:33

7. Demerits (lecture): it needs the CF, and the integrals can be very difficult. ESA ★★ T011V 00:06:31 · T011V 00:06:55 · T011Vb 00:04:19
8. Swapping y_1 and y_2 changes the sign of W but not the final PI. ESA ★★ T011Vb 00:09:02 · T011Vb 00:09:20

Why?

Why are u' and v' given by those formulas?
Try $y_p = uy_1 + vy_2$ and impose $u'y_1 + v'y_2 = 0$; then $y_p' = uy_1' + vy_2'$ and substituting into $y'' + py' + qy = X$ leaves $u'y_1' + v'y_2' = X$ (because y_1, y_2 solve the homogeneous equation). Cramer's rule on the two equations gives $u' = -\frac{y_2 X}{W}$, $v' = \frac{y_1 X}{W}$.
ADDED FOR CLARITY S011V·5

Why must the coefficient of y'' be 1?
The derivation substitutes into $y'' + py' + qy = X$; the second condition is $u'y_1' + v'y_2' = X$ only for that form. For $2y'' + \cdots = R$ the right-hand side to use is $X = \frac{R}{2}$.
ADDED FOR CLARITY S011V·6 · T011V 00:06:06

Why may constants of integration be dropped?
A constant k in u adds ky_1 to y_p, which is already in the CF c_1y_1; the same for v and y_2.
T011V 00:04:18 · S011V·5

Why is $W \neq 0$ needed?
W is the determinant of the system for u', v'; it is non-zero exactly because y_1, y_2 are linearly independent.
T011V 00:03:05 · S011V·5

Exam traps

Exam trap. Forgetting the minus sign in $u = -\int \frac{y_2 X}{W} dx$, or pairing y_1 with u's integral: u goes with y_2 inside, v with y_1. ADDED FOR CLARITY S011V·5

Exam trap. Sign of W: for e^{3x}, e^{-x} it is $-4e^{2x}$, negative. Carry the sign through u and v. S011V·8 · S011V·14

Exam trap. "Linearly independent if $c_1y_1 + c_2y_2 \neq 0$" (lecture) is loose: it means $c_1y_1 + c_2y_2 \equiv 0$ forces $c_1 = c_2 = 0$. ADDED FOR CLARITY T011V 00:02:06

2 Slide and lecture examples: xe^{-x}, $\tan 2x$, $\frac{2}{1+e^x}$, a Cauchy equation

Key points

1. $y'' - 2y' - 3y = xe^{-x}$ could be done by the Type 4 rule; by VoP $W = -4e^{2x}$, u by parts, $v = -\frac{x^2}{8}$; the two PIs differ by $-\frac{e^{-x}}{64}$, a CF term. ESA ★★ S011V·7 · S011V·10 · T011V 00:08:02
2. For $\tan 2x$ write $\frac{\sin^2 2x}{\cos 2x} = \sec 2x - \cos 2x$ before integrating. ESA ★★ S011V·12 · T011Vb 00:04:58
3. For $\frac{2}{1+e^x}$: partial fractions $\frac{1}{e^x(1+e^x)} = \frac{1}{e^x} - \frac{1}{1+e^x}$, then multiply by e^{-x} to get $\frac{e^{-x}}{1+e^{-x}}$, a $\frac{dt}{t}$ form. ESA ★★ S011V·15 · S011V·16 · T011Vb 00:11:06 · T011Vb 00:12:10
4. For a Cauchy equation, first put $x = e^z$; then do VoP in z and return to x at the end (lecture only). ESA ★ (TRANSCRIPT ONLY) T011Vb 00:17:57 · T011Vb 00:20:11 · T011Vb 00:23:01

Exam traps

Exam trap. Slide 011V·13's box writes $y_p = -\frac{1}{4}\log(\sec 2x + \tan 2x)$: the factor $\cos 2x$ is missing. The correct PI is $-\frac{1}{4}\cos 2x \log(\sec 2x + \tan 2x)$ (the slide's own previous line; CW P28). S011V·13 · CW U2 P28

Exam trap. In $(D^2 - 1)y = \frac{2}{1+e^x}$ the constant -1 in y_p is not a CF term (the CF is $c_1e^x + c_2e^{-x}$): do not drop it. (ADDED FOR CLARITY) S011V·17

3 Trigonometric right-hand sides: $\sec ax$, $\csc x$, $e^x \tan x$, $e^{-x}\sec^3 x$

Key points

1. With complex roots and $X = e^{\alpha x}g$, the exponentials cancel: $W = \beta e^{2\alpha x}$ (E3.1). ESA Jan-24 s1 3a and s2 3a are this pattern. ESA ★★★ ADDED FOR CLARITY ESA Jan-24 s1 3a · ESA Jan-24 s2 3a · CW U2 P25
2. Know the integrals: $\int \tan = \log\sec$, $\int \cot = \log\sin$, $\int \sec = \log(\sec+\tan)$, $\int \frac{\sin}{\cos^3} = \frac{\sec^2}{2}$. ESA ★★★ CW U2 P23 · CW U2 P26 · ESA Jan-24 s2 3a · S011V·12

4 Repeated roots with $X = \frac{e^{mx}}{x^k}$, and the leading-coefficient trap

Key points

1. ESA Jul-23 3c = Jun-24 3a and May-24 3c are the same pattern: $(D-m)^2 y = \frac{e^{mx}}{x}$, answer $y = (c_1 + c_2 x)e^{mx} + xe^{mx}\log x$. ESA ★★★ ESA Jul-23 3c · ESA Jun-24 3a · ESA May-24 3c
2. State it when you absorb a term: "$-xe^{mx}$ is a multiple of y_2 and is absorbed into c_2". ESA ★★ ADDED FOR CLARITY ESA Jul-23 3c · CW U2 P24

Why?

Why can the operator rules not do $\frac{e^{-3x}}{x}$?
Type 4 gives $e^{-3x}\frac{1}{D^2}\frac{1}{x}$, i.e. integrating $\frac{1}{x}$ twice, which is possible ($x\log x - x$) — the VoP route just organises the same integrals. The question asks for VoP, so use it.
ADDED FOR CLARITY ESA Jul-23 3c

Exam traps

Exam trap. CW P27 ($2y'' - y' - \cdots = 25e^{-x}$): divide by 2 first, so $X = \frac{25}{2}e^{-x}$, not $25e^{-x}$. CW U2 P27 · S011V·6

Exam trap. "Solve" means $y = y_c + y_p$: the ESA questions want the general solution, not only y_p.
ADDED FOR CLARITY ESA Jul-23 3c · ESA May-24 3c

★ Past-paper questions

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024 (19 distinct Unit 2 questions, 24 occurrences). Variation of parameters is asked in 4 distinct questions, 5 occurrences, every time worth 6–7 marks and every time with the words "by the method of variation of parameters": Jul-23 3c = Jun-24 3a ($y''+6y'+9y=\frac{e^{-3x}}{x}$), May-24 3c ($y''-2y'+y=\frac{e^{x}}{x}$), Jan-24 s1 3a ($(D^2-2D+2)y=e^{x}\tan x$, = CW P25) and Jan-24 s2 3a ($y''+2y'+2y=e^{-x}\sec^3 x$). It appears in 5 of the 8 papers, so the method, the Wronskian and the u, v formulas are ★★★; the $\frac{e^{mx}}{x}$ repeated-root pattern (2 distinct, 3 occurrences) and $e^{\alpha x}\times$trig ($\tan$, $\sec^3$) are ★★★; the slide-only right-hand sides (xe^{-x}, $\frac{2}{1+e^{x}}$) and the CW-only ones ($\sec ax$, $\csc x$, $\tan 2x$) are ★★; the use on variable-coefficient equations (lecture only, never asked) is ★.

3 · Trigonometric right-hand sides: $\sec ax$, $\csc x$, $e^{x}\tan x$, $e^{-x}\sec^3 x$

ESA Jan 2024 set 1 (7 marks) UE23MA141A Q3.a · solved in W3.3

Solve $(D^2-2D+2)y=e^{x}\tan x$ by the method of variation of parameters (= CW P25).

ESA Jan 2024 set 2 (7 marks) UE23MA141A Q3.a · solved in W3.4

Solve $y''+2y'+2y=e^{-x}\sec^3 x$ by the method of variation of parameters.

4 · Repeated roots with $X=\frac{e^{mx}}{x^{k}}$, and the leading-coefficient trap

ESA Jul 2023 (6 marks) UE22MA141A Q3.c · solved in W4.1

Solve $y''+6y'+9y=\frac{e^{-3x}}{x}$ by the method of variation of parameters.

ESA Jun 2024 (6 marks) UE23MA141A Q3.a · solved in W4.1

Solve $y''+6y'+9y=\frac{e^{-3x}}{x}$ by the method of variation of parameters (repeat of Jul-23 3c).

ESA May 2024 (6 marks) UE22MA141A Q3.c · solved in W4.2

Solve $\frac{d^2y}{dx^2}-2\frac{dy}{dx}+y=\frac{e^{x}}{x}$ by the method of variation of parameters.